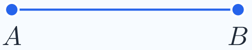
Segment AB

$\{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$

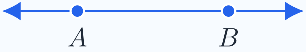


Ray $\overrightarrow{AB}$

Union of AB and points X where $\mathcal{B}(A, B, X)$

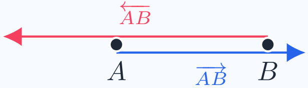
Line $\overleftrightarrow{AB}$

Extends infinitely in both directions



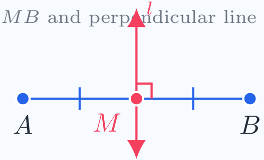
Intersection of Opposite Rays

$\overleftarrow{AB} \cap \overrightarrow{AB} = AB$



Midpoint & Bisector

$AM \cong MB$ and perpendicular line $l \perp AB$



Parallelism ($l \parallel m$)

No points in common: $l \cap m = \emptyset$